

End Semester Examination May 2025

Name of the course: B. Tech

Semester: VIII

Name of the Paper: Mobile Ad hoc Networks

Paper Code: TEC 857

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two questions among a, b, & c in each main question.
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain the following i) IrDA ii) HomeRF	CO1
(b)	Explain the basic operating principle of wireless ad hoc networks.	
(c)	How ad hoc networks is useful in military and emergency applications.	
Q2	(20 marks)	
(a)	Explain the "Hidden Terminal Problem" and "Exposed Terminal Problem" in ad hoc wireless networks. How these issues are reduced by MACA protocol.	CO1/CO2
(b)	Explain the following MAC protocols with Flow Chart: i) DBTMA ii) MACA- by- Invitation	
(c)	With reference to the Bluetooth, illustrate its operational states in detail. Also explain the trade-off of synchronization and bandwidth utilization in 'Scatternet'..	
Q3	(20 marks)	
(a)	Compare the Intrazone (IARP) and Interzone (IERP) routing protocol. Illustrate the IERP operation with example.	CO3
(b)	Explain the properties of AODV routing protocol. Discuss the distinguish features of DSDV and AODV routing protocols.	
(c)	Explain the Route Discovery mechanism of DSR routing protocol. How the piggyback of small data packet is beneficial to avoid infinite recursion of route discoveries.	
Q4	(20 marks)	
(a)	What are the design issues of routing protocols. Explain the ideal characteristics of routing protocol.	CO3/ CO4
(b)	What are the effects of beaconing of battery life? Explain smart batteries and battery characteristics.	
(c)	Explain the complete working of 'Associativity Based Routing' (ABR).	
Q5	(20 marks)	
(a)	Explain the wireless sensor network architecture. What is 'Rumor Routing'? Explain it with example.	CO5
(b)	Explain the following network layer attacks: i) Wormhole attack ii) Byzantine attack	
(c)	Describe the issues and challenges in providing QoS in ad hoc wireless networks.	

